

A pilot study of yogic meditation for family dementia caregivers with depressive symptoms: Effects on mental health, cognition, and telomerase activity

BACKGROUND This study examined the effects of brief daily yogic meditation on mental health, cognitive functioning, and immune cell telomerase activity in family dementia caregivers with mild depressive symptoms. **METHODS** Thirty-nine family dementia caregivers (mean age 60.3 years old (SD=10.2)) were randomized to practicing Kirtan Kriya or listening to relaxation music for 12 minutes per day for eight weeks. The severity of depressive symptoms, mental and cognitive functioning were assessed at baseline and follow-up. Telomerase activity in peripheral blood mononuclear cells (PMBC) was examined in peripheral PBMC pre- and post-intervention. **RESULTS** The meditation group showed significantly lower levels of depressive symptoms and greater improvement in mental health and cognitive functioning compared to the relaxation group. In the meditation group, 65.2% showed 50% improvement on the Hamilton Depression Rating scale and 52% of the participants showed 50% improvement on the Mental Health Composite Summary score (MCS) of the SF-36 scale; compared to 31.2% and 19% respectively in the relaxation group ($p < 0.05$). The meditation group showed 43% improvement in telomerase activity compared to 3.7% in the relaxation group ($p = 0.05$). **CONCLUSION** This pilot study found that brief daily meditation practices by family dementia caregivers can lead to improved mental and cognitive functioning, and lower levels of depressive symptoms. This improvement is accompanied by an increase in telomerase activity suggesting improvement in stress-induced cellular aging. These results need to be confirmed in a larger sample.